

Static Members and Encapsulation

Q1: Static Variable

```
class Counter {
    static int count = 0;
    Counter() {
        count++;
    }

    public static void main(String[] args) {
        new Counter();
        new Counter();
        new Counter();
        System.out.println("Count: " + count);
    }
}
```

What will be the output? Explain why.

Q2: Static Block Execution

```
class StaticBlockExample {
    static int x;

    static {
        x = 10;
        System.out.println("Static block executed");
    }

    public static void main(String[] args) {
        System.out.println("Value of x: " + x);
    }
}
```

What gets printed first and why?

Q3: Static Method Access

```
class Demo {  
    static int data = 50;  
  
    static void display() {  
        System.out.println("Data: " + data);  
    }  
  
    public static void main(String[] args) {  
        Demo.display();  
    }  
}
```

Q: Can the `display()` method access `data`? Why or why not?

Q4: Mixing Static and Non-Static

```
class Example {  
    static int x = 10;  
    int y = 5;  
  
    void show() {  
        System.out.println("x: " + x + ", y: " + y);  
    }  
  
    public static void main(String[] args) {  
        Example e = new Example();  
        e.show();  
    }  
}
```

Q5: Create a Program

Task: Write a Java program with:

- a static variable `collegeName`
- a non-static variable `studentName`
- a constructor that takes the student name
- a static method to change the college name for all

Q6: Basic Encapsulation

Create a class `Account` with:

- private fields `accountNumber`, `balance`
- public getters and setters
- validation in setter to ensure `balance` cannot be negative

Q: Try setting a negative balance. What happens?

Q7: Encapsulation with Constructor

Create a class `Employee` with:

- private fields `name`, `salary`
- constructor to initialize the values
- method to display the employee details

Q: Can these fields be accessed directly in `main`?

Q8: Modify with Validation

Modify the above `Employee` class so that:

- `salary` cannot be less than 10,000
- If such a value is passed, set default 10,000

Q9: Combine Concepts

Create a class `LibraryBook`:

- private fields: `title`, `author`
- static field: `libraryName`
- constructor to initialize title/author
- static method to change library name
- method to display book details

Create multiple books, change the library name and display the result.